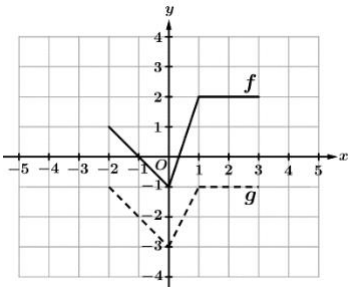
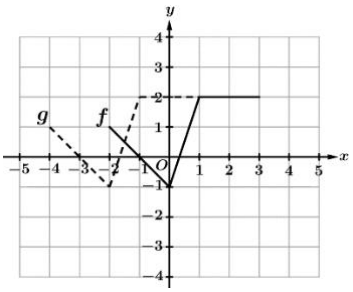
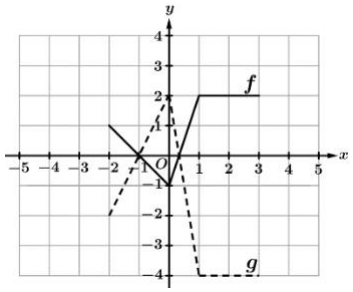
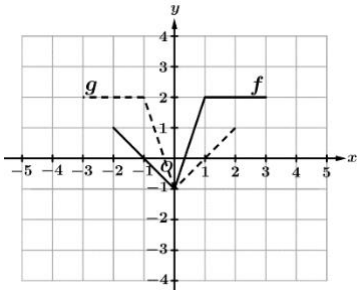
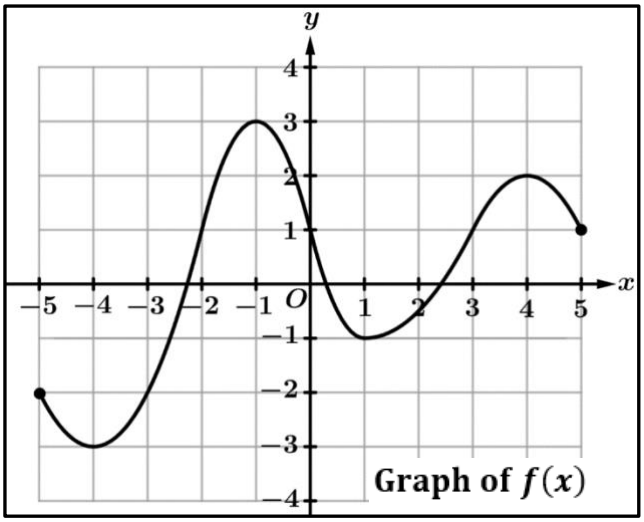
Transformations of Functions		
Transformation	**Geometric Result**	**Example**
$g(x) = f(x) + k$	f has a vertical translation of k units	 $g(x) = f(x) - 2$
$g(x) = f(x + h)$	f has a horizontal translation of $-h$ units	 $g(x) = f(x + 2)$
$g(x) = af(x)$	f has a vertical dilation by a factor of $\|a\|$ **Note**: If $a < 0$, f is reflected over the x-axis.	 $g(x) = -2f(x)$
$g(x) = f(bx)$	f has a horizontal dilation by a factor of $\left\|\frac{1}{b}\right\|$ **Note**: If $b < 0$, f is reflected over the y-axis.	 $g(x) = f(-x)$



The graph of f is shown in the figure above. Use the graph of f to answer the following.

Example 1: Find the domain and range of the function $f(x)$.

Domain: _____ **Range:** _____

Example 2: Let $g(x) = 2f(x-1) + 3$.

a) Find the domain and range of g.

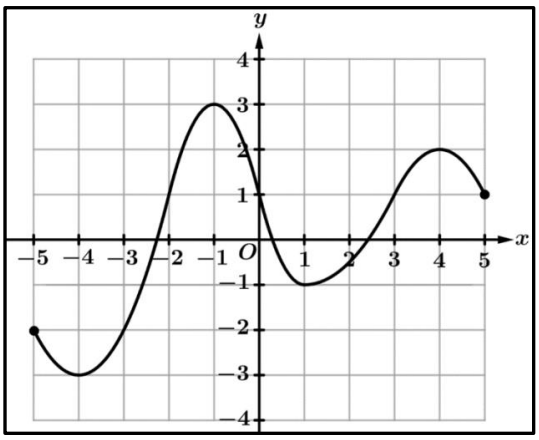
Domain: _____ **Range:** _____

b) The point $(4, 2)$ on $f(x)$ transforms to which point on the graph of $g(x)$?

c) Find $g(0)$.

d) How many solutions does the equation $g(x) = 0$ have? Explain your reasoning.

e) On what intervals is $g(x)$ increasing?



Example 3: Let $h(x) = -2f(x+3) - 1$.

a) Find the domain and range of h.

Domain: _____ **Range:** _____

a) On what intervals is the graph of h concave up?

b) On what intervals is h increasing?

c) Find any solutions to the equation $h(x) = 5$.

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x	0	1	2	4	8
$m(x)$		3		5	-12

x	0	1	2	4	8
$n(x)$	-4	0			13

Example 4: The partially completed tables of values above represent the functions $m(x)$ and $n(x)$, where $n(x) = -3m(2x) + 5$. Find the missing values in the tables above.

More Fun with Transformations

$$f(x) = -2(x+3)^2(x-1)$$

The function f is given above. Use the function f to answer the following problems.

Example 5: Let $g(x) = f(-x)$

a) Find the zeros of $g(x)$.

b) On what intervals is $g < 0$?

Example 6: Let $h(x) = 2f(3x)$.

a) Find $h(1)$.

b) On what intervals is $h \geq 0$?

c) What is the y-intercept of h?

$$r(x) = \frac{(x-3)(x+1)}{(x-3)(x+4)}$$

The rational function r is given above. Use the function r to answer the following problems.

Example 7: Let $k(x) = -4r\left(-\frac{x}{2}\right)$

a) Identify any x-values where the graph of k has a vertical asymptote or a hole.

b) For which values of x is $k(x) \geq 0$?

x	-2	0	2	4	6
$p(x)$	1	-1	0	3	7

The function p is a cubic polynomial with selected values given in the table above.

Example 8: Let $m(x) = -3p(x + 4)$

a) Describe the transformations of p that lead to the function m.

b) Name an x-value where the graph of $m(x)$ has a zero.

c) Find the average rate of change of $m(x)$ over the interval $[0,\ 2]$.

d) Find the domain and range of $m(x)$.

e) For which x-value does $m(x) = 3$?

The domain and range of the function $n(x)$ are $[-6,\ 9]$ and $[-8,\ 10)$, respectively. Additionally, n is increasing on the intervals $[-6,\ 1]$ and $[5,\ 9)$ and decreasing on the interval $[1,\ 5]$.

Example 9: Let $q(x) = 4 - 2n(5 - x)$.

a) Find the domain and range of $q(x)$.

b) On what intervals is q increasing and decreasing?